Customer Shopping Behavior Analysis

Data Analytics Project Using Python, SQL, and Power BI

1. Introduction

Understanding customer purchasing behavior is critical for businesses that want to improve sales performance and customer retention. With the growth of e-commerce, organizations increasingly rely on data analytics to identify purchasing patterns, evaluate product performance, and optimize marketing strategies.

In this project, I analyzed a customer shopping behavior dataset using **Python, SQL, and Power BI**. The goal was to transform raw transactional data into meaningful insights that can help businesses make data-driven decisions.

The analysis focuses on answering several key questions such as:

- Which customer groups generate the most revenue?
- Do subscribers spend more than non-subscribers?
- Which products receive the highest customer ratings?
- How do discounts influence purchasing behavior?
- What patterns exist across different customer segments?

By combining data processing, database querying, and interactive dashboards, the project demonstrates a complete **end-to-end data analytics workflow**.

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2. Project Workflow

The project follows a structured analytics pipeline commonly used in industry.

Step 1 — Data Collection
Customer shopping behavior data was provided in CSV format containing transactional and demographic information.

Step 2 — Data Preparation (Python)
Python was used to inspect the dataset, verify column types, and ensure the data was ready for analysis.

Step 3 — Data Analysis (SQL)
The cleaned dataset was loaded into a relational database where SQL queries were used to analyze revenue, customer behavior, and product performance.

Step 4 — Data Visualization (Power BI)
Key metrics and insights were visualized using an interactive Power BI dashboard to support business decision-making.

This workflow mirrors the process used by data teams in many organizations.

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3. Dataset Overview

The dataset contains several thousand customer transactions and includes attributes describing customer demographics, product purchases, and transaction details.

Some of the most relevant variables include:

Column	Description
Customer ID	Unique identifier for each customer
Age	Customer age
Gender	Customer gender
Item Purchased	Product purchased
Category	Product category
Purchase Amount	Value of the transaction
Location	Customer location
Review Rating	Customer rating for purchased item
Subscription Status	Whether the customer has an active subscription
Discount Applied	Indicates if a discount was used
Shipping Type	Delivery option selected
Previous Purchases	Number of past purchases made by the customer

Column	Description
Payment Method	Method used to complete the purchase

These variables allow analysis from both a **customer perspective** and a **product performance perspective**.

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4. Data Preparation Using Python

Python was used as the initial step to explore and prepare the dataset.

The preparation stage included:

- Loading the dataset using **Pandas**
- Inspecting column names and data types
- Checking for missing or inconsistent values
- Preparing the dataset for database storage

	age	age_group
0	55	Middle_age
1	19	Young
2	50	Middle_age
3	21	Young
4	45	Middle_age
5	46	Middle_age
6	63	Senior_citizen
7	27	Young
8	26	Young
9	57	Middle_age

To better understand customer demographics, the raw **age variable** was transformed into meaningful **age groups**. Customers were categorized into three segments: **Young, Middle_age, and Senior_citizen**. This transformation helps simplify demographic

analysis and enables easier comparison of purchasing behavior across different age segments.

For example, customers in their early twenties are categorized as **Young**, while customers in their mid-40s to late-50s fall under the **Middle_age** segment. Customers above a certain threshold are classified as **Senior_citizen**.

Creating age groups allows businesses to analyze how purchasing patterns vary across different life stages. This type of segmentation is commonly used in marketing analytics to design **targeted campaigns, personalized offers, and customer engagement strategies**.

```
#creating purchase_frequence_days column

frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}

df['purchase_frequence_days'] = df['frequency_of_purchases'].map(frequency_mapping)


df[['purchase_frequence_days','frequency_of_purchases']].head(10)
```

In the dataset, the **frequency of purchases** column originally contained categorical values such as *Weekly*, *Monthly*, *Quarterly*, and *Annually*. To make this information usable for quantitative analysis, these categories were converted into **numeric values representing the number of days between purchases**.

A mapping dictionary was created in Python to assign each frequency category a corresponding numeric value. For example:

Weekly → 7 days
Fortnightly / Bi-Weekly → 14 days
Monthly → 30 days
Quarterly / Every 3 Months → 90 days
Annually → 365 days

Using the Pandas **map() function**, a new column called **purchase_frequency_days** was generated. This transformation allows analysts to perform statistical calculations, build predictive models, and analyze customer purchasing behavior more effectively.

Converting categorical frequency data into numeric values makes it easier to measure **customer engagement, purchase cycles, and long-term purchasing trends**.

Once validated, the dataset was loaded into a SQL database where further analysis could be performed efficiently.

Using Python for preprocessing helps ensure that the data is **clean, structured, and reliable** before performing analytical queries.

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5. SQL Analysis

SQL was used to perform analytical queries that answer practical business questions.

Some of the key analyses included:

- Calculating total revenue by customer demographics
- Identifying customers who used discounts but still made high-value purchases
- Determining top-rated products based on review ratings
- Comparing spending behavior between subscribers and non-subscribers
- Evaluating the impact of different shipping options
- Segmenting customers based on purchase history

For example, the following query calculates total revenue generated by each gender:

```
SELECT gender, SUM(purchase_amount)
FROM customer_data
GROUP BY gender;
```

	gender	sum(purchase_amount)
▶	Male	157890
	Female	75191

Customer segmentation was implemented using a **Common Table Expression (CTE)** and conditional logic:

```
WITH customer AS (
SELECT customer_id, previous_purchases,
CASE
WHEN previous_purchases = 1 THEN 'New'
WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
ELSE 'Loyal'
END AS customer_segment
FROM customer_data
)
SELECT customer_segment, COUNT(*)
FROM customer
GROUP BY customer_segment;
```

customer_segment	n of customers
Loyal	3116
Returning	701
New	83

These SQL queries allow the dataset to be analyzed from multiple perspectives and provide insights relevant to business operations.

Key Insights from the Analysis

Several interesting patterns emerged during the analysis.

Customer Segmentation

Customers were categorized into three groups based on the number of previous purchases:

- **New Customers**
- **Returning Customers**
- **Loyal Customers**

The analysis showed that a large portion of transactions comes from **loyal customers**, indicating strong repeat purchasing behavior. This suggests that customer retention strategies are already effective and should continue to be prioritized.

❖ **Subscription Impact**

The analysis compared spending patterns between customers with and without a subscription.

While both groups show similar average purchase values, subscribers contribute a significant portion of total revenue due to their frequent purchases. This indicates that subscription programs play an important role in maintaining consistent sales.

- **Product Ratings**

Products with higher review ratings tend to receive more positive customer feedback and may have stronger long-term sales potential. Identifying these products helps businesses focus marketing efforts on items that already have strong customer satisfaction.

- **Discount Usage**

Discounts were widely used across multiple product categories. The analysis showed that certain products have a high percentage of purchases involving discounts, indicating that promotional pricing can influence customer purchasing decisions.

This insight may help businesses design **targeted promotional campaigns** to increase sales.

- **Shipping Preferences**

Customers used multiple shipping options including standard and express delivery. The average purchase value across shipping types did not vary significantly, suggesting that shipping choice may be influenced more by convenience than spending behavior.

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6. Power BI Dashboard

To present the analysis in a clear and interactive format, a Power BI dashboard was developed.

The dashboard includes several visual components:

Sales Overview

- Total revenue
- Distribution of purchases across product categories
- Seasonal purchasing patterns

Customer Analysis

- Revenue by gender
- Age group spending trends
- Customer segmentation

Product Insights

- Top rated products
- Most frequently purchased items
- Discount usage patterns

The dashboard allows users to interact with filters and explore insights dynamically, making it useful for **business decision-makers and analysts**.

Business Recommendations

Based on the analysis, several recommendations can be made:

- **Strengthen Loyalty Programs**
  Since loyal customers generate a large share of purchases, businesses should continue investing in loyalty rewards and personalized offers.

- **Promote Highly Rated Products**
  Products with strong customer ratings should be highlighted in marketing campaigns, as they already demonstrate positive customer satisfaction.
- **Use Discounts Strategically**
  Discounts appear to influence purchasing behavior in certain categories. Businesses should use targeted promotions rather than applying discounts across all products.
- **Expand Subscription Benefits**
  Encouraging more customers to join subscription programs could help increase repeat purchases and stabilize revenue.

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7. Conclusion

This project demonstrates how customer transaction data can be transformed into actionable insights using modern analytics tools.

By combining **Python for data preparation, SQL for analysis, and Power BI for visualization**, the project illustrates a complete analytics workflow that can support business decision-making.

Understanding customer purchasing patterns, product performance, and promotional effectiveness allows businesses to improve marketing strategies, increase customer retention, and optimize overall sales performance.